



Introduction to Bioengineering Theory (BBL1020; 3-0-0)

Lecture-5: 2 Classes



Please Listen Carefully

&

Keep your phone(s) on silent mode



.....is the basis of life



Chemistry is the basis of life

Chemical basis of life



“The origin of life presents a particularly difficult problem for chemistry because it is essentially unconstrained: there are a very, very large number of reactions that might have occurred on pre biotic Earth, and few hard limits on the types of compounds that might have been available, on the possible reaction conditions, and on the processes that might finally have led to elementary cells of the type that we now know.”

From Whitesides Research Group, Harvard University

“What is, however, clear is the central conundrum of understanding “life”: that is, that a living cell is composed of molecules; that molecules in cells are reacting; that neither the molecules nor the reactions are individually alive; but that the cell – as a collection of reactions and processes - is alive.

Several theories of the origins of life propose that life emerged spontaneously from the self-assembly of organic reactions, (im)probably occurring chaotically in complex mixtures of molecules. The chemistry that permit the assembly of simple organic reactions into networks with complex emergent behaviors, however, remain in completely understood.

Conditions were different from today

Different temperatures, types of molecules, number of molecules, pressure, charge density, □ possible of reactions was vast □ small molecules □ complex molecules □



Prof. George M. Whitesides
Department of Chemistry and
Chemical Biology
Harvard University

Chemical basis of Life

Chemicals play many roles in an organism life:

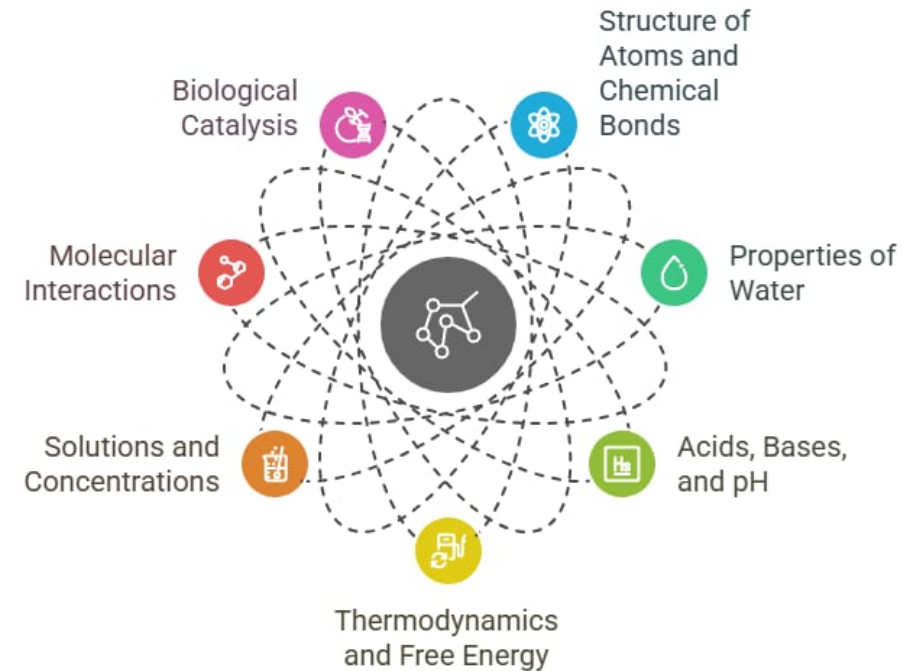
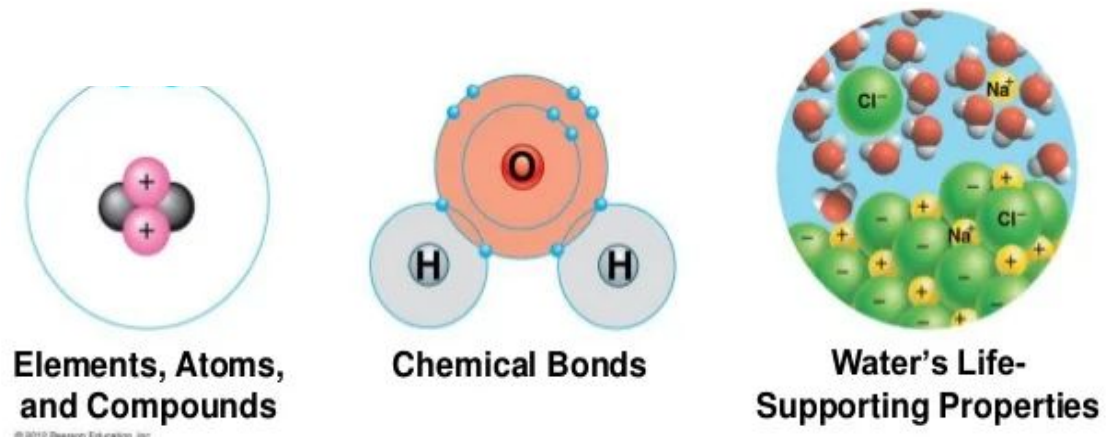
1. Organisms use chemical signals for **muscle movements, mating and defense.**
2. DNA is a combination of chemical elements
3. Chemical reactions of body occur in cells consisting of 70-95% water.

Organism are composed of elements



Element	% by mole
Hydrogen	63.0
Oxygen	26.0
Carbon	9.0
Nitrogen	1.25
Calcium	0.25
Phosphorus	0.19
Potassium	0.06
Sulfur	0.06
Sodium	0.04
Chlorine	0.025
Magnesium	0.013
Iron	0.00004
Iodine	0.000002

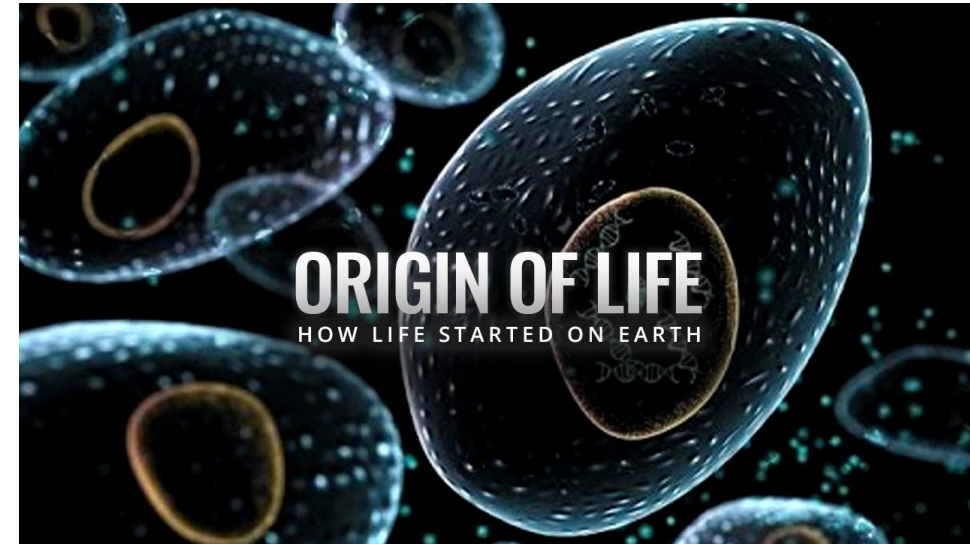
99% of atoms in a human body come from these 4 elements



But in the beginning, there were **no** proteins:

So how did cells evolve to create them?

And how did the first cell evolve in the first place-- and cross the border from chemistry into the new, living world of biology?



Origin and Evolution of Life

How?

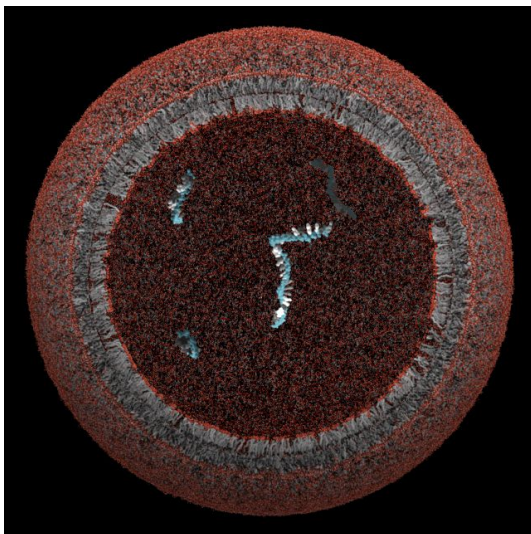
- **Panspermia Theory**
- **RNA World Theory**

This theory proposed that life developed elsewhere in the universe and then arrived on earth. Microbial spores from space were supposedly brought to earth by asteroids or meteors; which jump-started life on earth. However, this theory states **how life spread, not how life began**. The validity of this theory is fiercely debated since no conclusive proof exists. Furthermore, this theory ignores the fact that space is incredibly vast, so travelling between planets would take thousands and thousands of years. It would be highly improbable for a lifeform to remain dormant or alive for such long durations.

When? Originated about 3.7 billion years ago



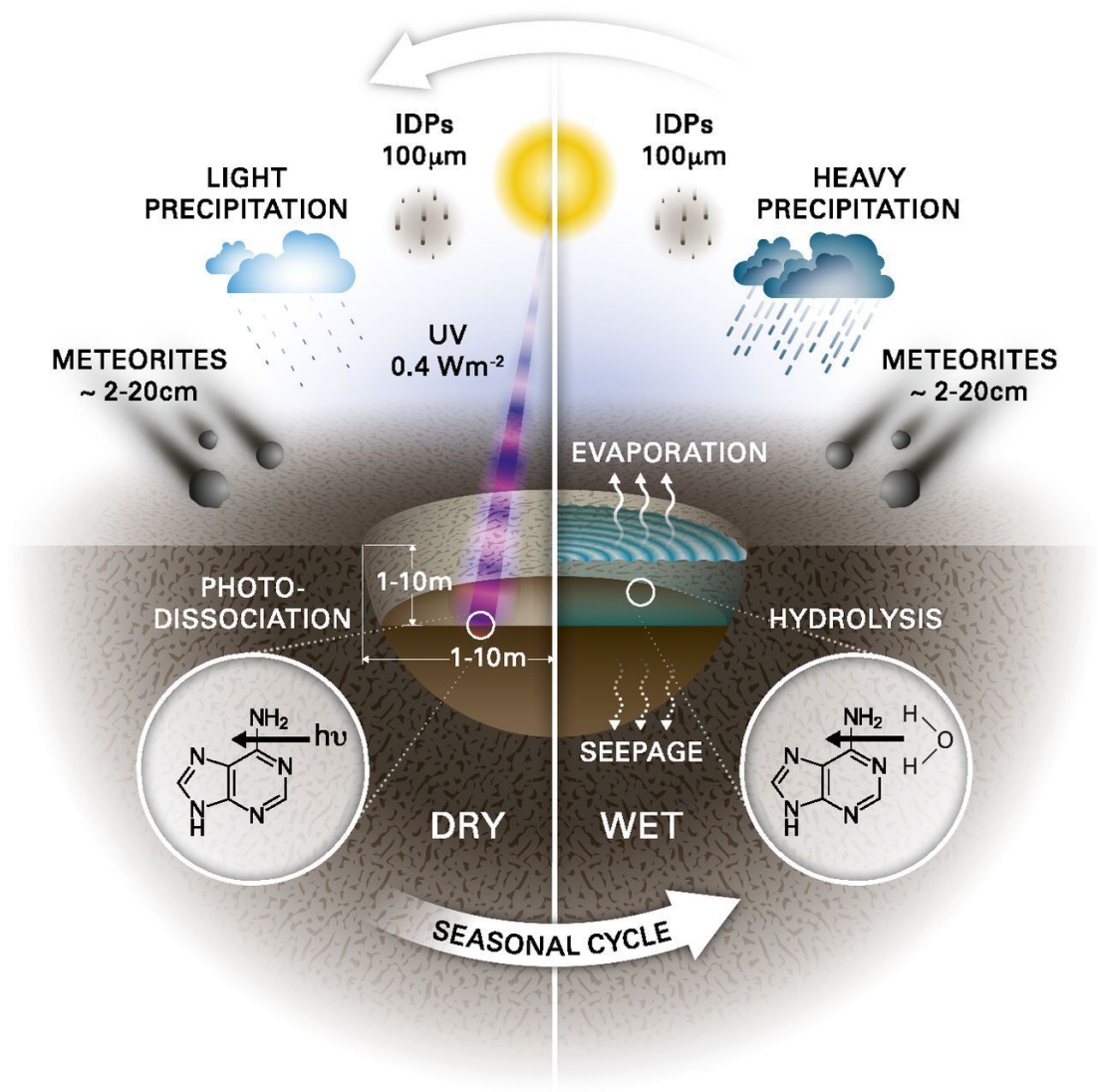
<https://www.youtube.com/watch?v=X1MBuUvddf8>



Nobel laureate **Jack Szostak** researches the origin of life by recreating primitive cells, or "**protocells**," in the lab, focusing on how simple molecules formed membranes and self-replicating RNA, proposing early life began in geothermal lake environments with fluctuating temperatures (**hot/cold cycles**) that aided RNA replication and vesicle division, leading to Darwinian evolution from simple fatty acid vesicles encapsulating genetic material.

He aims to bridge the gap between non-living chemistry and early biology, suggesting the first **life was a simple, self-sustaining system that grew, divided, and evolved.**

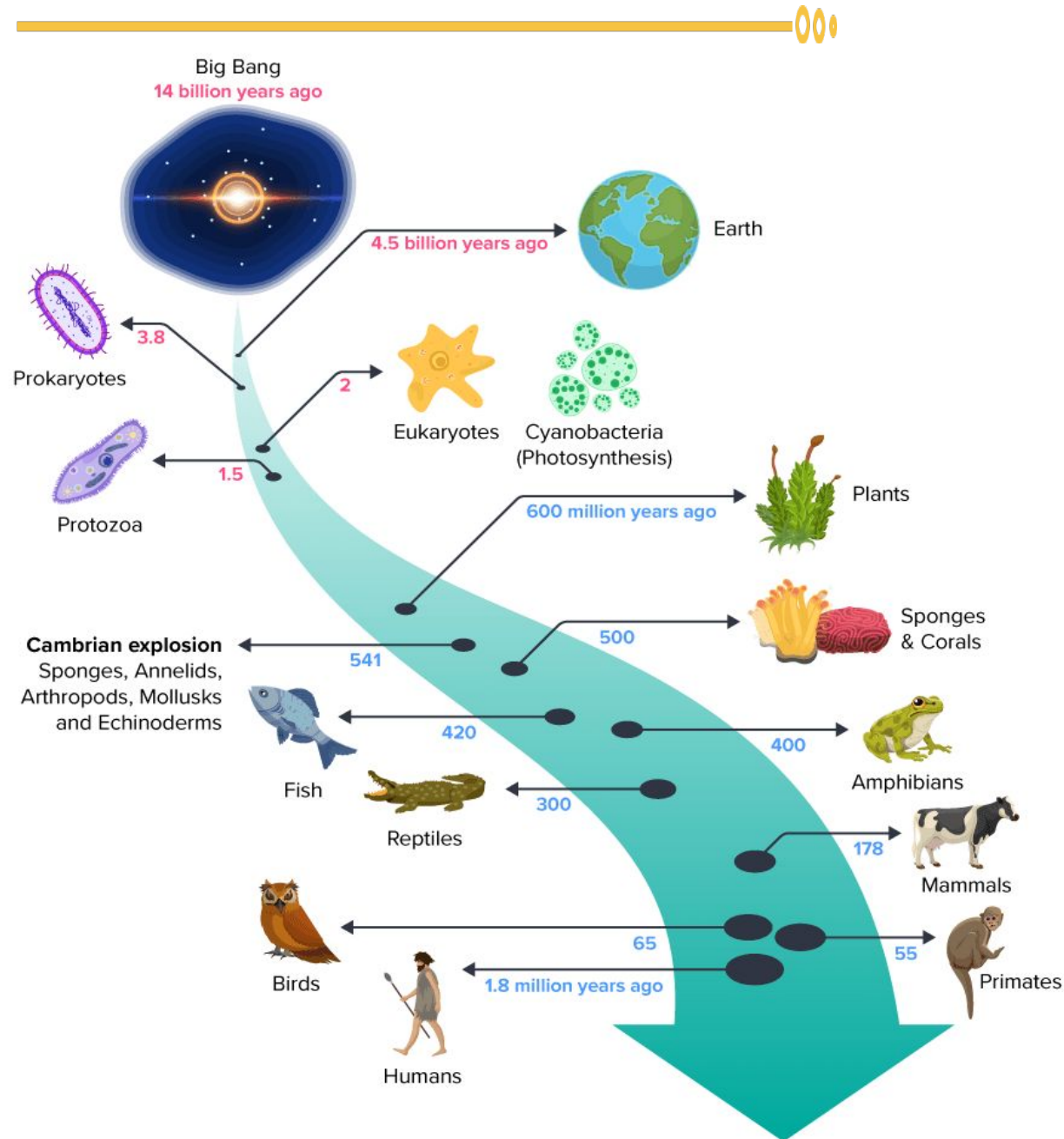
Origin and Evolution of Life



RNA World

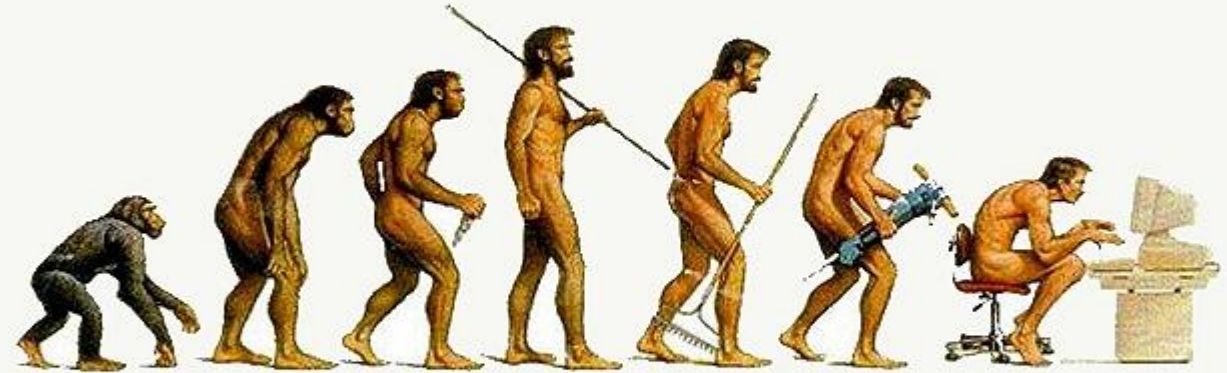


Origin and Evolution of Life



Darwin's theory of evolution

Evolution



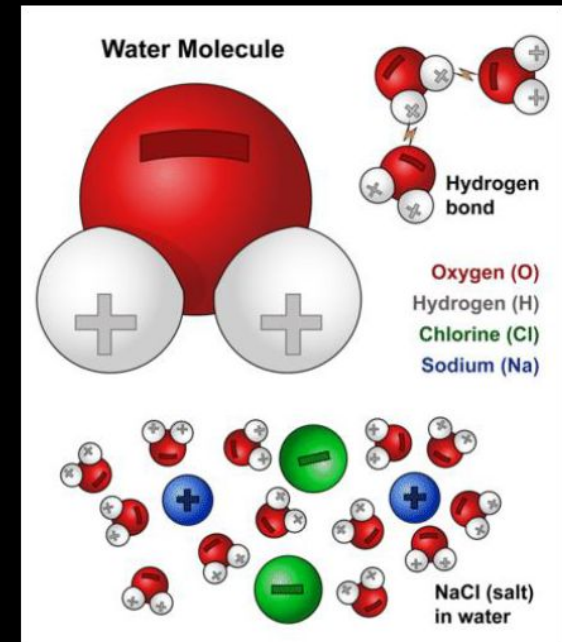
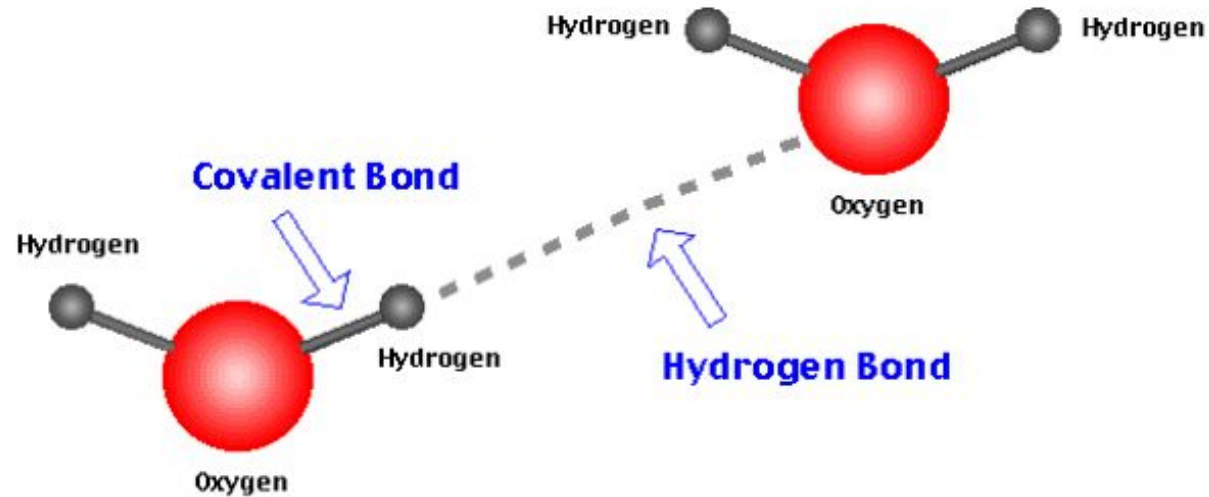
(OR is it?)

- Natural selection
- Evolution
- The origin of species

Water sustains life



A simple small molecule yet governs all form of the life



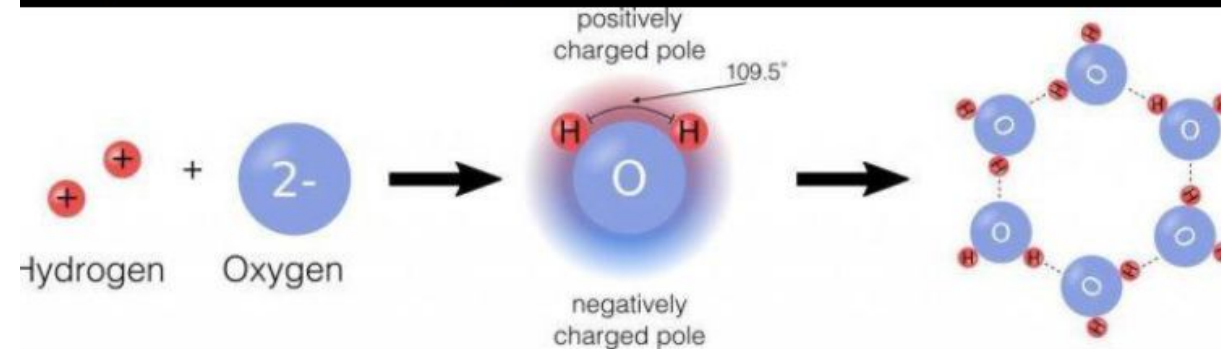
Molecule which can be present naturally in three distinct phases : Solid, Liquid and Gas.

Water likes polar molecules and hates non-polars (hydrophobics)

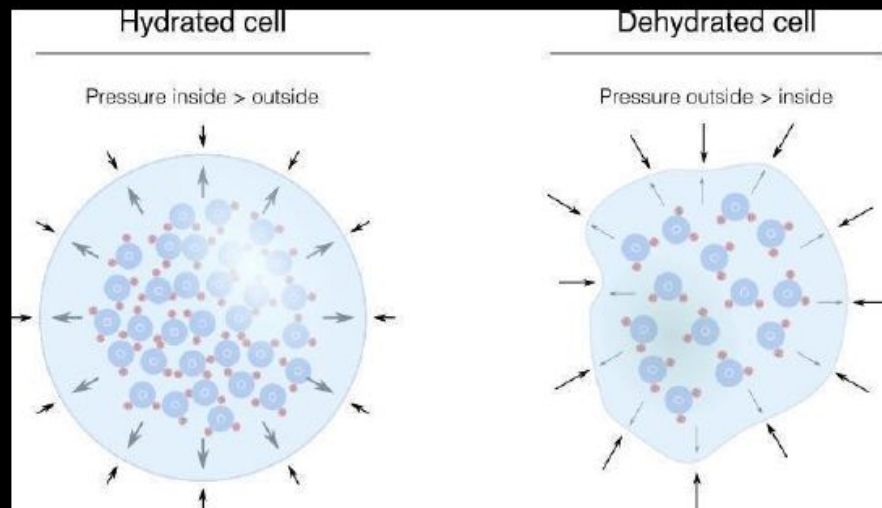
Biological role of water



1. Molecular make-up of water: asymmetric and polar

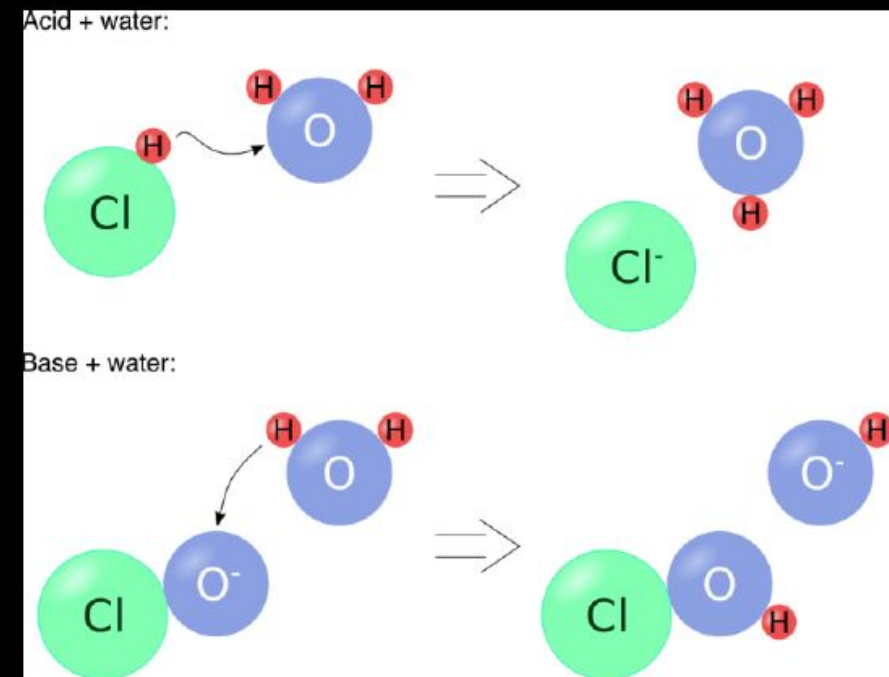


2. Supports cellular structure



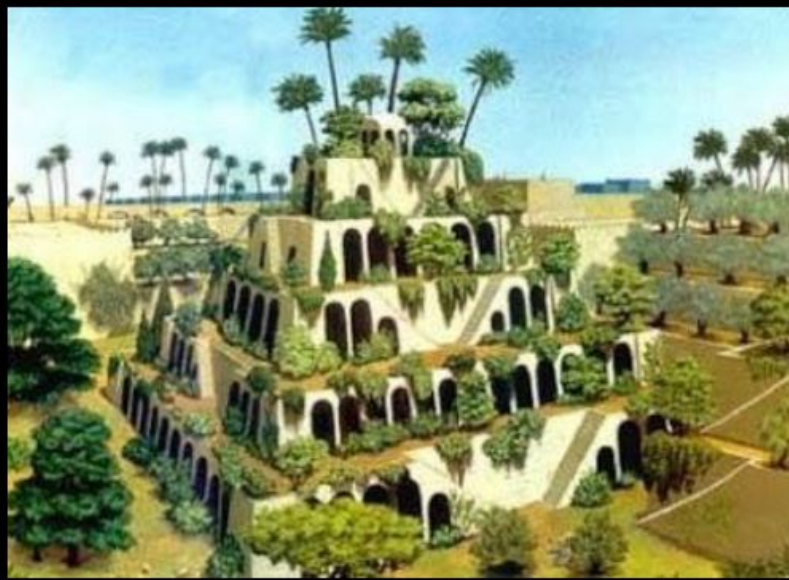
3. Universal solvent: cohesion, molecules and ions interactions

4. Facilitates chemical reactions: buffer

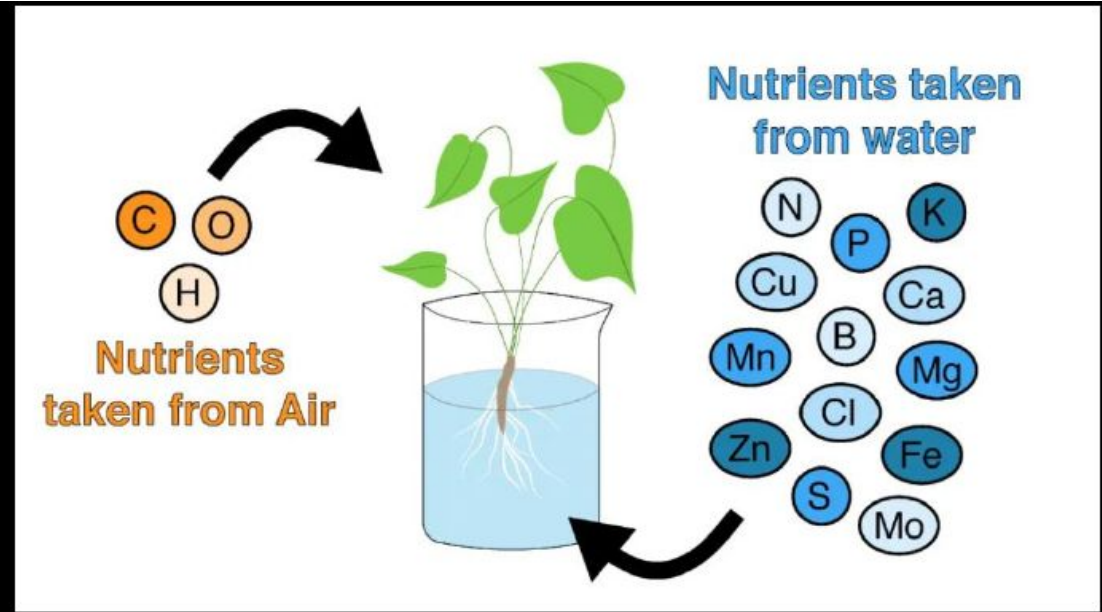


Hydroponics: The power of water to grow food

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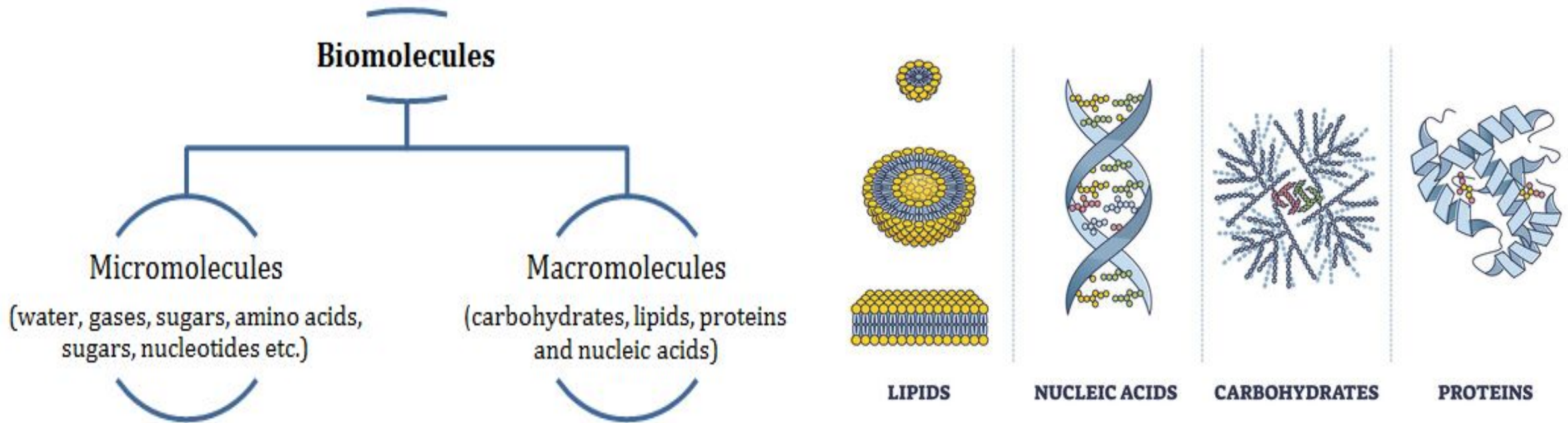
pulley-system of
water from the river



Vertical Farming

Biomolecules

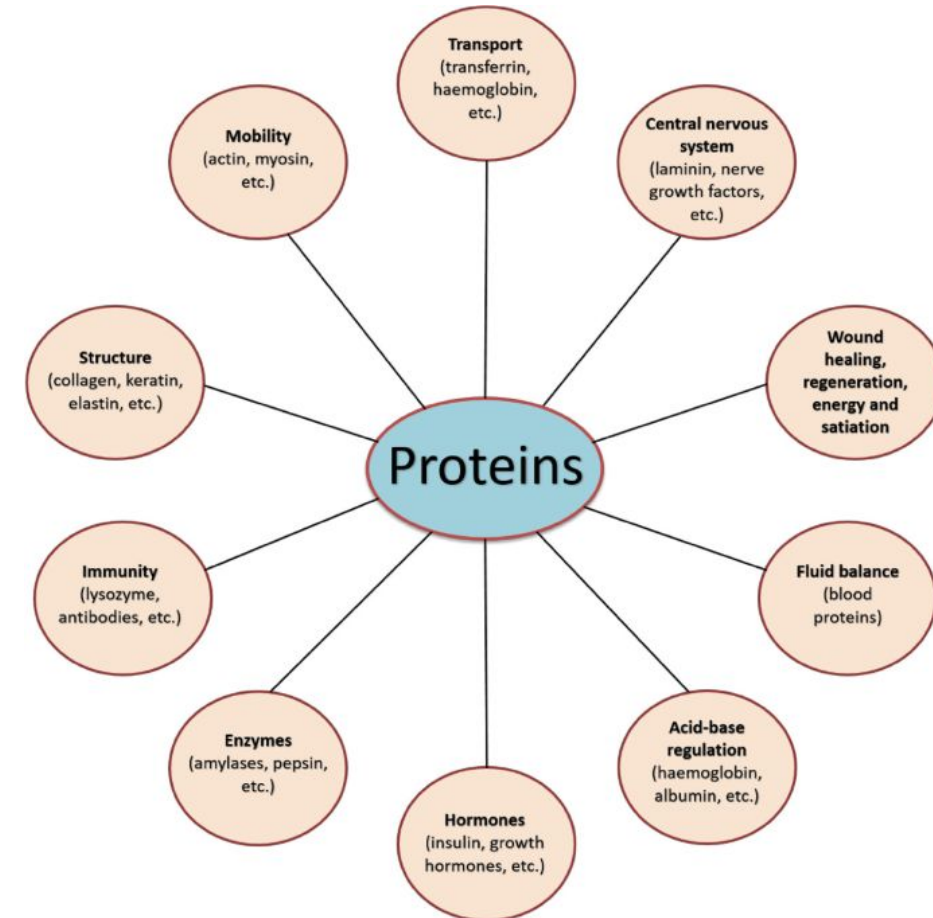
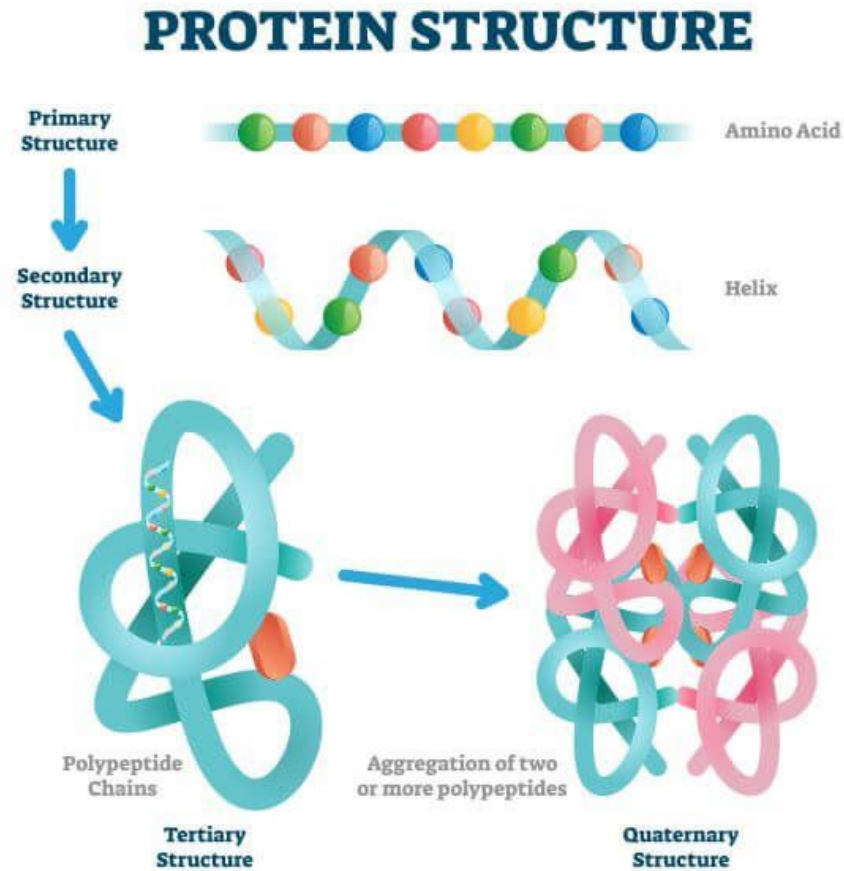
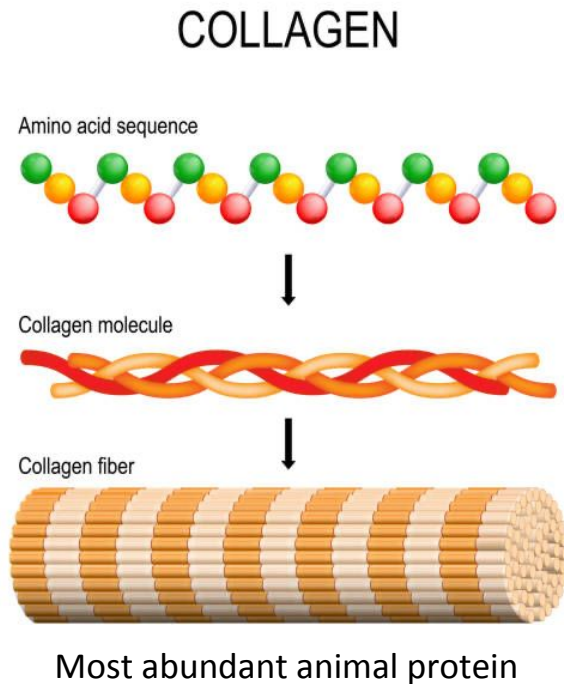
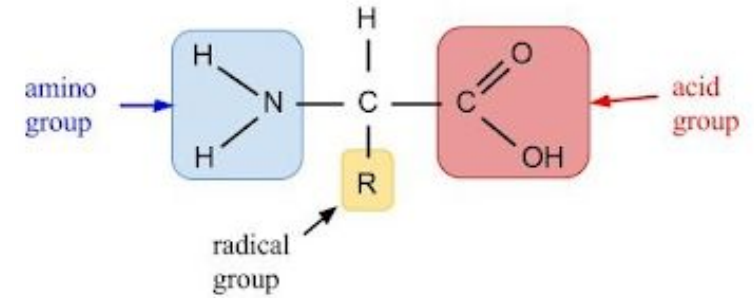
Chemical or Molecules present in the living organisms are called as Biomolecules. They include organic and inorganic compounds.



A molecule having a **relatively low molecular weight** is classified as Micromolecule, whereas large molecules (**>1000Da**) usually made up of polymerization of smaller molecules are classified as Macromolecules.

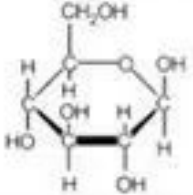
Biological macromolecules

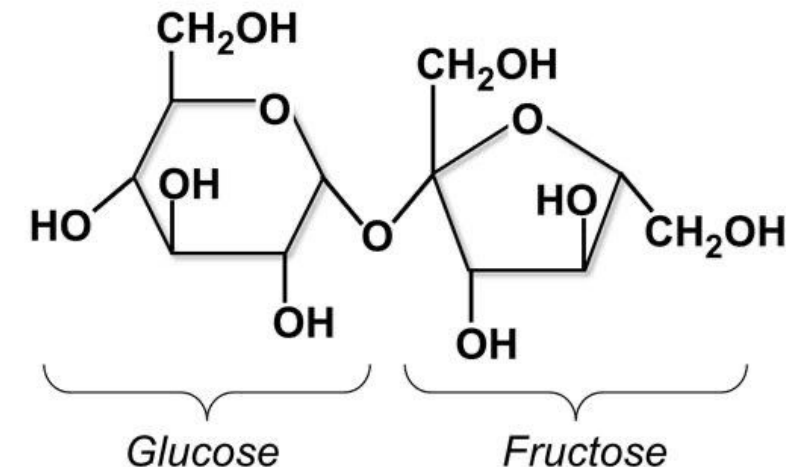
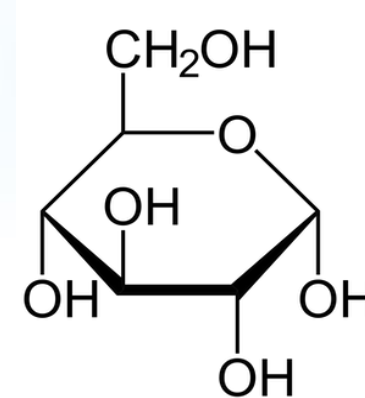
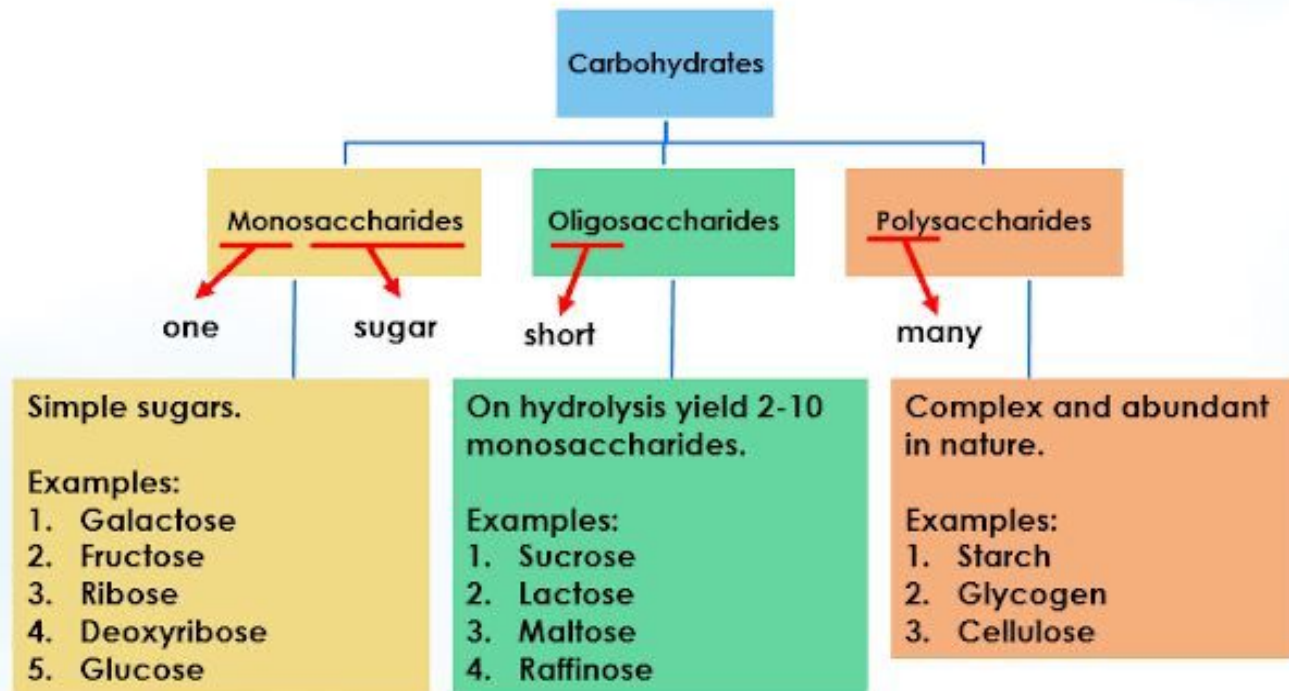
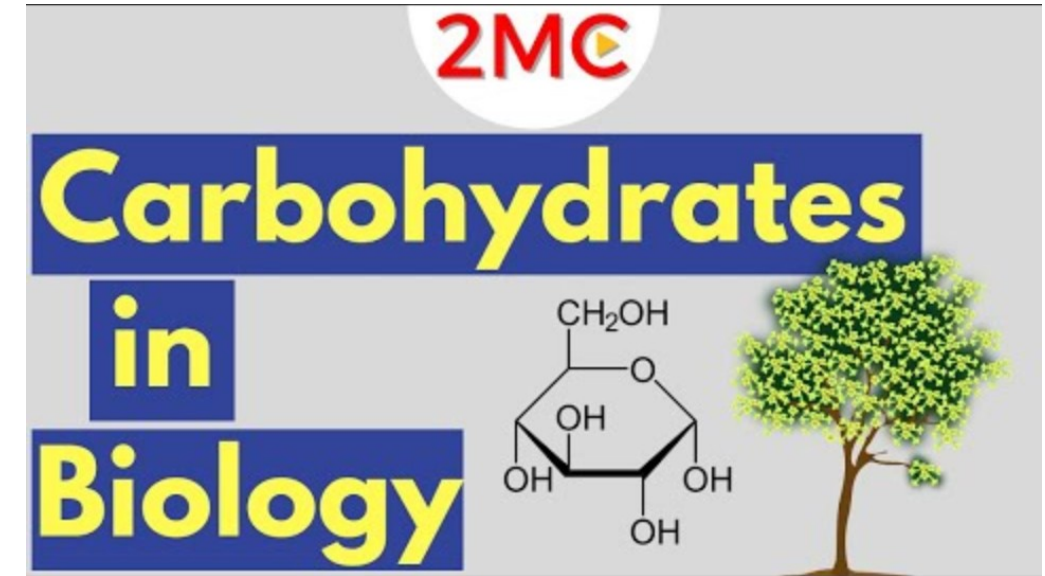
- Proteins:** Proteins are the heteropolymers of amino acids, linked together by peptide bond. There are 20 standard amino acids, found in the structure of proteins.



Functions of Protein

2. **Carbohydrates**: These are the most abundant organic molecules in nature and are the **primary source of energy**.

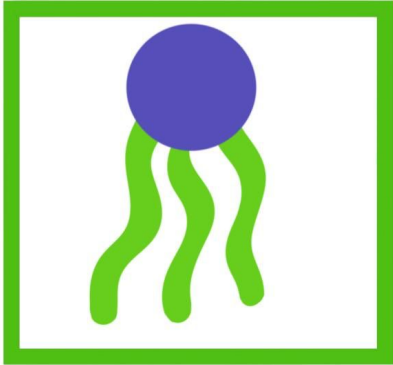
Biomolecules	Monomer(s)	Function	Food Source	Example(s)
Carbohydrates 	Monosaccharides	Provide material to build cell membrane Quick energy for cells	Pastas Breads Fruits Vegetables	Glucose Fructose Lactose Cellulose



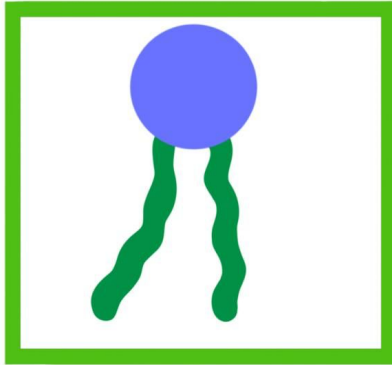
3. **Lipids** Lipids are made up of fatty acid monomers which is responsible for **energy storage**, **cell structural integrity** and **cellular signalling**.

Main types of lipids:

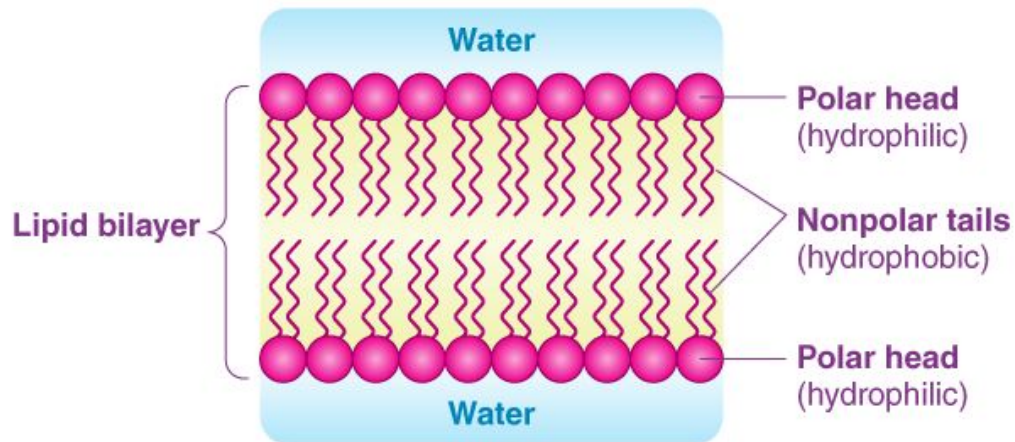
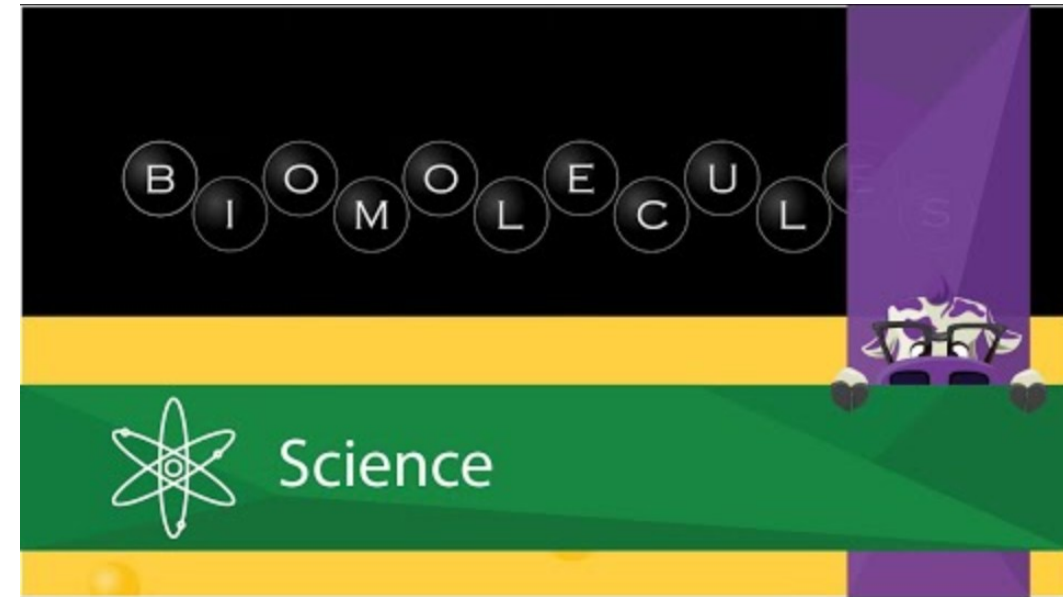
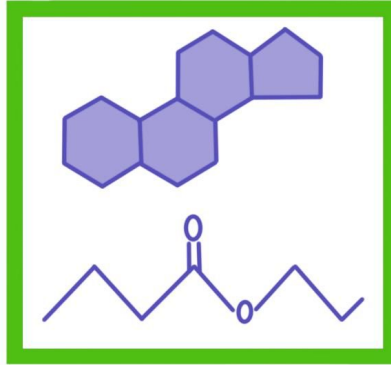
Triglycerides



Phospholipids

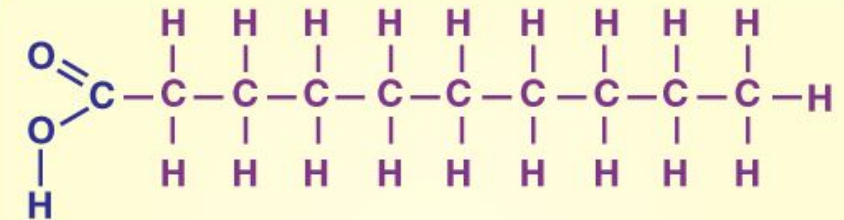


Steroids and Waxes

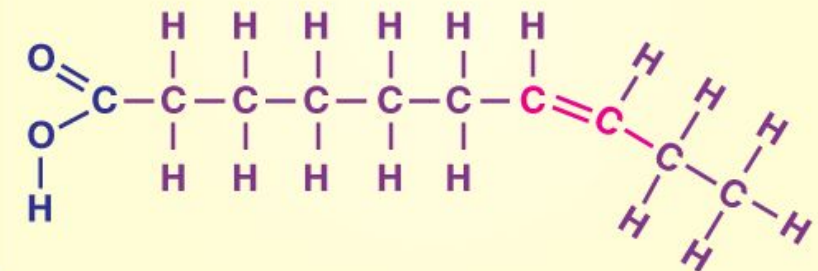


Phospholipids are responsible for cell membrane structure

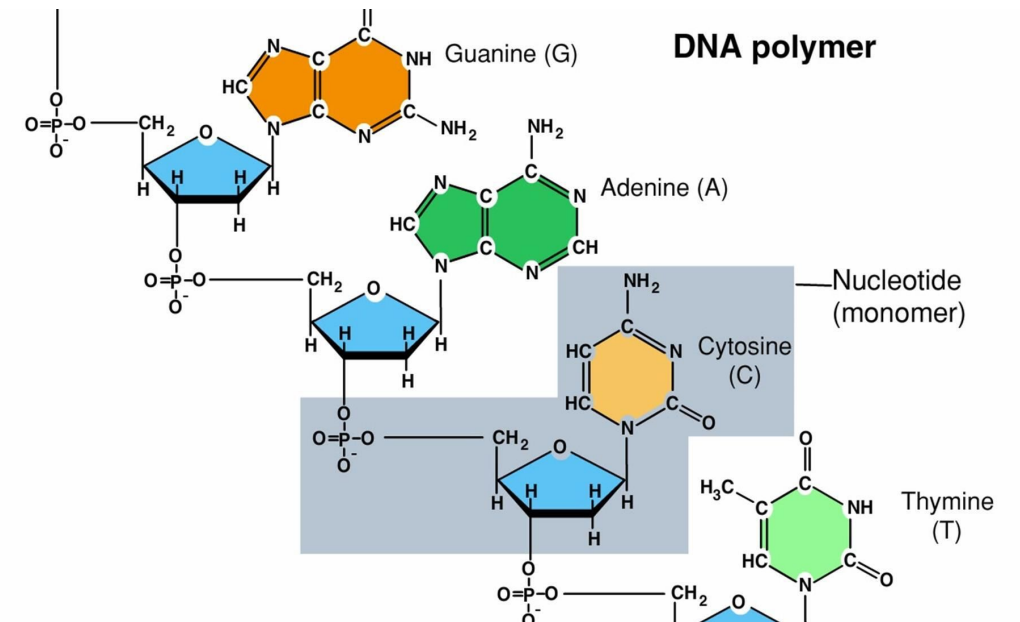
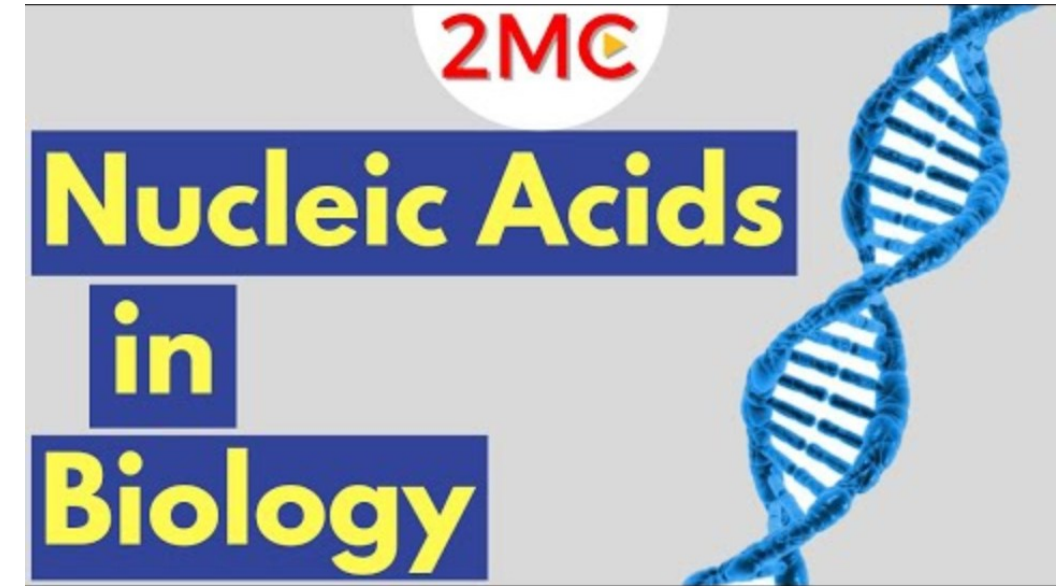
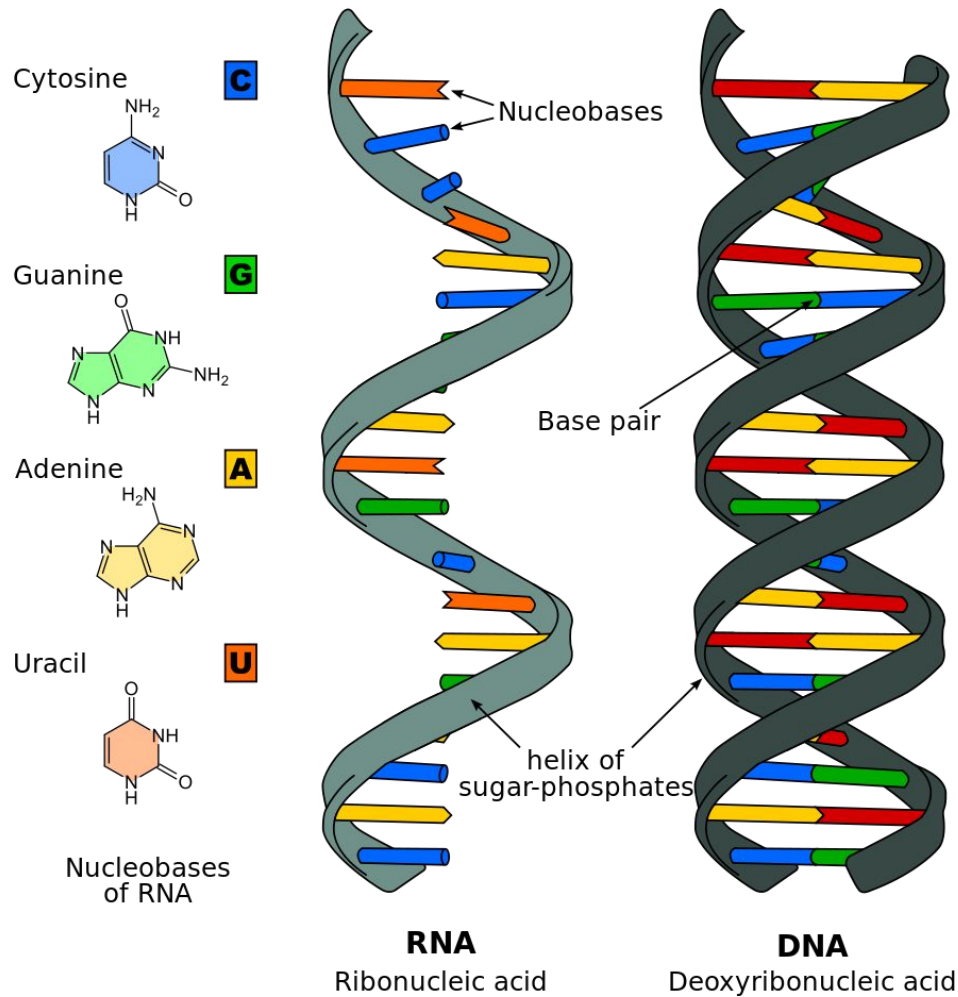
(a) Saturated



(b) Unsaturated



4. **Nucleic Acids**: Nucleic acids are made up of **pentose sugar**, **nitrogenous base** and **phosphate group**.



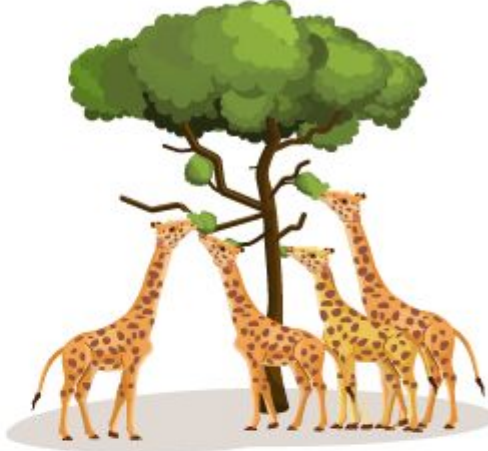
Evolution

Evolution takes place over millions of years, gradually resulting in new species

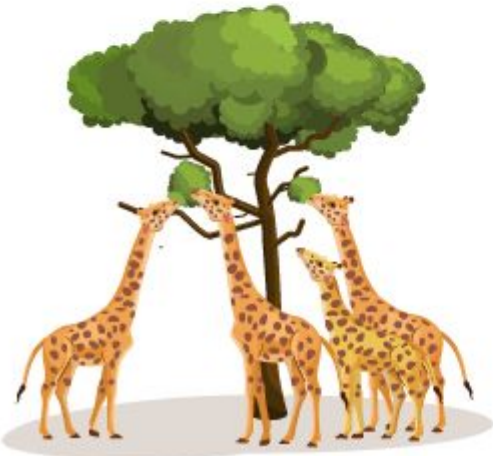
1. Variation in neck length



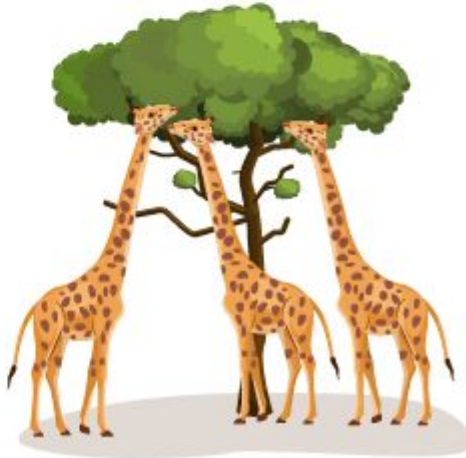
2. Struggle for existence



3. Greater fitness



4. Long-neck trait increases



It is not the strongest that survive, nor the most intelligent, but the ones most responsive to change.

- Charles Darwin

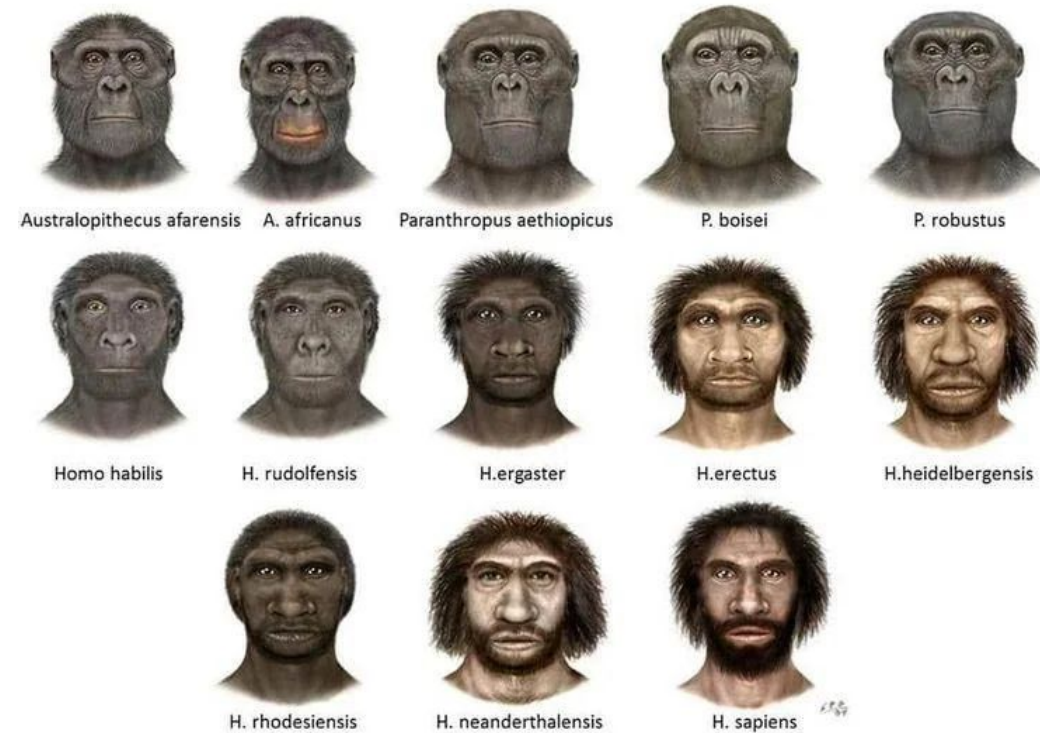


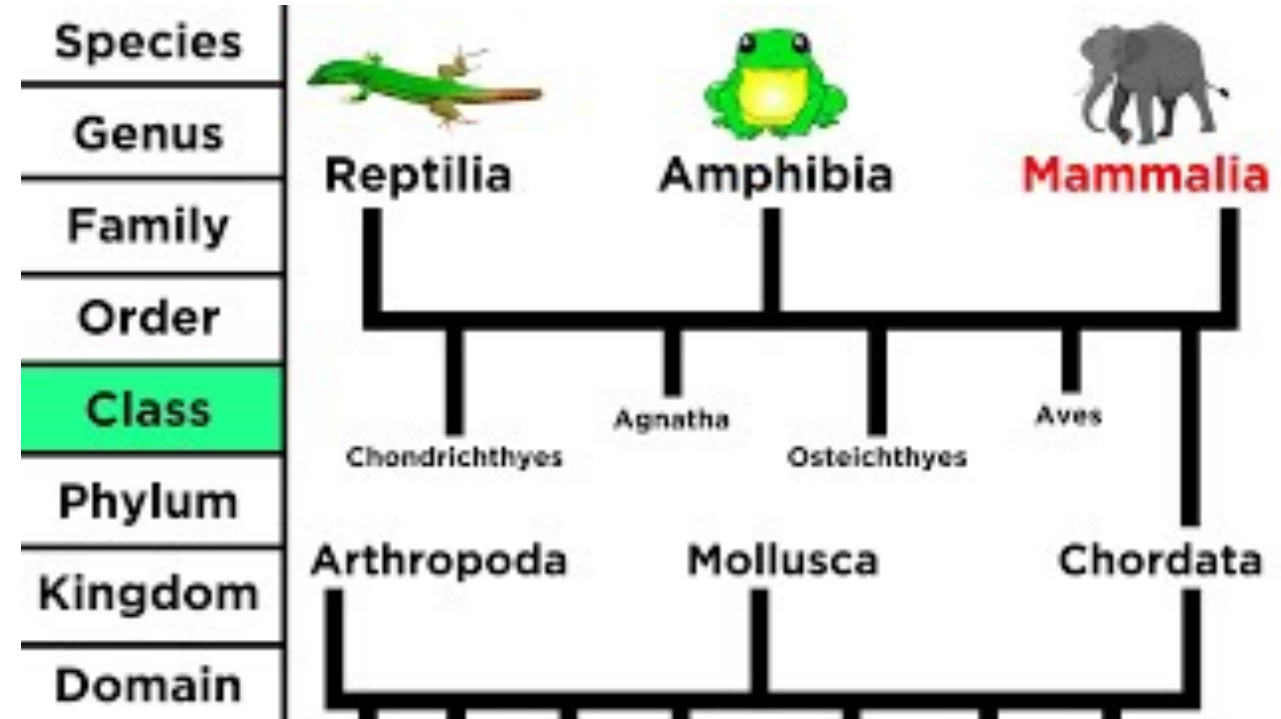
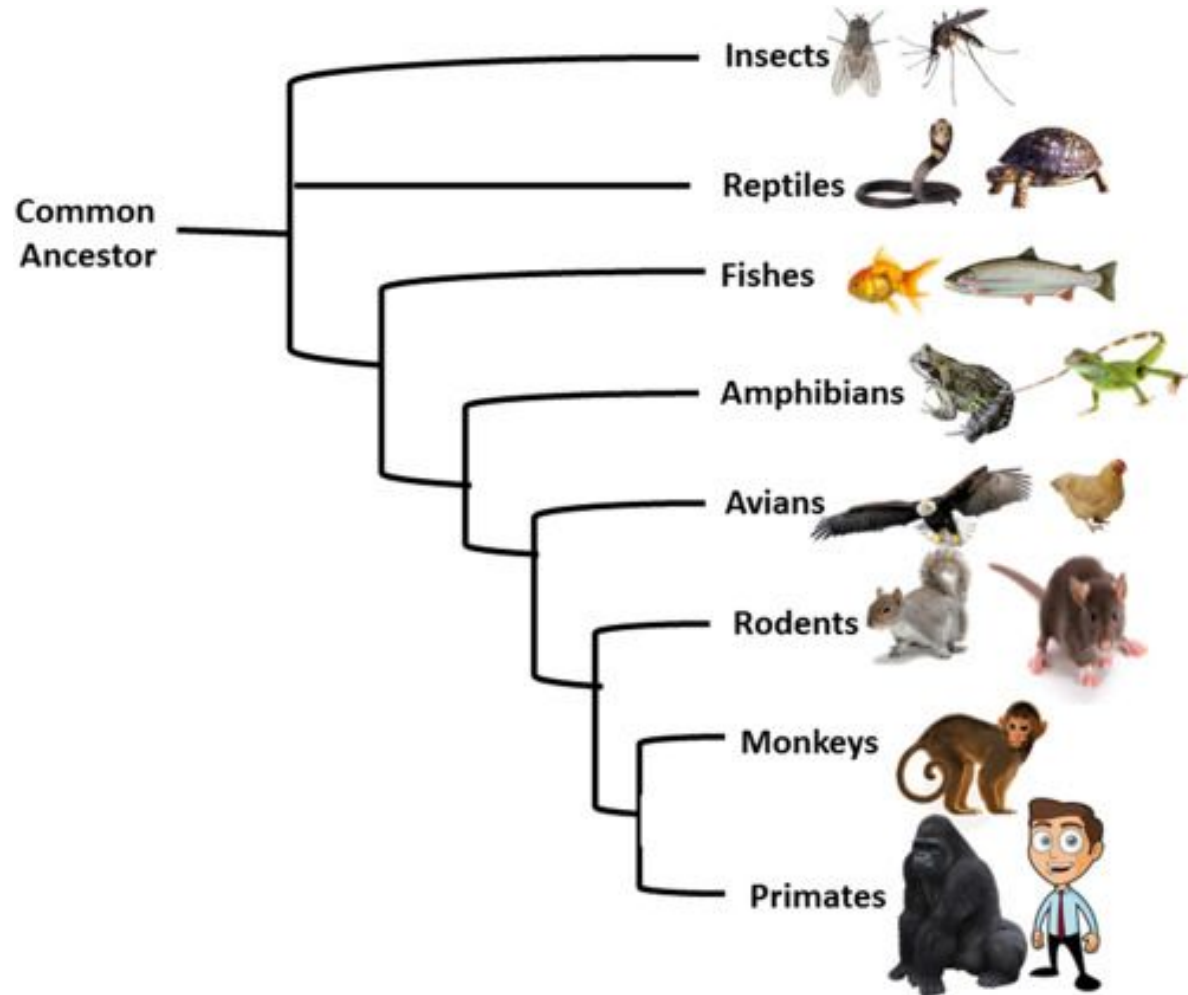
1.2 Million species of plants and animals have been identified

Scientists **estimate** that the total number of species in the world is 8.7 million

Evolution

Is the reason we have biodiversity in the world.





Phylogeny helps us to understand the importance of each species in the world.

Extinction

Extinct species are the ones which were considered “**NOT FIT**” by nature

Although nowadays Human intervention is the main cause of extinction. Convention on biological diversity stated that almost 150 species are going extinct daily.

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Extinct Bird Re-Evolved Itself Back Into Existence

Once the bird became extinct, it took only 20,000 years for it to return and evolve into a flightless bird again.

By [Alexa Lardieri](#) | May 10, 2019, at 3:37 p.m.

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What Caused the extinction of dinosaurs? What do you think?

Is It Possible?